Corporate Finance

Lecture 8: Capital Structure in a Perfect Market (II)

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Plan

- Recap from lecture 7
- Ask how much leverage raises the cost of equity: Modigliani-Miller Proposition II
- See why the weighted average cost of capital does not move
- Write the same result with betas, and use it to price projects from comparables
- Dismantle two popular arguments for debt: earnings per share and dilution
- Practice on three exercises

Mostly based on the BDM textbook Chap. 14.3-14.5.

Recap lecture 7

Recap of Lecture 7

Let's do a recap:

- The assumptions of MM I, and what "no taxes" does to the WACC
- Tour moche SA financed with equity only: what the firm is worth
- The same project with 500 of debt, step by step: the value did not change
- The 543 fallacy: cheap debt, dearer equity, same average
- MM Proposition I
- Homemade leverage, and how it proves MM I with the law of one price
- The market value balance sheet
- And from Lecture 6: the unlevered cost of capital

Recap: the assumptions of MM I

- Modigliani and Miller (1958) work in a **perfect capital market** (Chap. 14.2):
  1. investors and firms trade the same securities at **fair prices**, equal to the present value of their cash flows;
  2. **no taxes**, transaction costs or issuance costs;
  3. **no bankruptcy costs**;
  4. all investors can **borrow and lend without restrictions at the risk-free rate**;
  5. financing changes neither the **cash flows** of the investments nor the **information** about them.
- Remove any one of them and the conclusions below can change. That is where Lectures 9 to 11 go.

Recap: no taxes, so the WACC loses its tax term

- In Lecture 6 the WACC carried a tax term on the debt, because interest is tax deductible:

$$r_{WACC} = \frac{E}{E+D}\, r_E + \frac{D}{E+D}\, r_D\,(1-\tau_C)$$

- Assumption 2 sets the corporate tax rate τ_C to zero, so $(1-\tau_C) = 1$ and the term disappears:

$$r_{WACC} = \frac{E}{E+D}\, r_E + \frac{D}{E+D}\, r_D$$

- This is the **pre-tax WACC**: the plain weighted average of what shareholders and lenders require. Debt has no tax advantage here.

So the only question left in Lectures 7 and 8 is whether cheap debt lowers this average **on its own**. Lecture 7 said no with an example. Today we say why with a formula.

Recap: Tour moche SA with equity only

Invest **800** today; receive **1400** (strong economy) or **900** (weak), each with probability one half. Cost of equity 15%, given (Chap. 14.1).

- With equity only, the cost of capital **is** the cost of equity: 15%.
- Expected cash flow $0.5 \times 1400 + 0.5 \times 900 = 1150$.
- Value of the firm $= 1150/1.15 = 1000$. NPV $= -800 + 1000 = 200$.
- The entrepreneur sells the whole equity for 1000, pays the 800 and keeps the 200. This equity is called **unlevered equity**; by construction its shareholders earn 15% on average.

Recap: the same project with 500 of debt, step by step

- **Step 1, the lenders.** The firm borrows 500 at the risk-free 5%, because even the weak outcome (900) repays the loan. Next year it owes $500 \times 1.05 = 525$ in both states.
- **Step 2, the shareholders.** Lenders are paid first; shareholders take what is left: $1400 - 525 = 875$ or $900 - 525 = 375$. This is **levered equity**.
- **Step 3, add them up.** $525 + 875 = 1400$ and $525 + 375 = 900$: debt and equity together receive **exactly the project's cash flows**.

Levered firm	Year 0	Strong	Weak
Debt	500	525	525
Levered equity	?	875	375
Firm	?	**1400**	**900**

Recap: step by step (con't): the value did not change

- **Step 4, the law of one price.** Two investments that pay the same cash flows in every state must sell for the same price. Debt plus equity pays what the unlevered firm pays, so the levered firm is also worth **1000**.
- **Step 5, the equity.** $E = 1000 - 500 = 500$.

Levered firm	Year 0	Strong	Weak
Debt	500	525	525
Levered equity	**500**	875	375
Firm	**1000**	1400	900

- **Step 6, the entrepreneur.** She raises $500 + 500 = 1000$ either way, pays 800 and keeps 200 either way. She is **indifferent** between the two financings.

Changing the mix of debt and equity moved money between lenders and shareholders. It did not change the size of the pie.

Recap: the 543 fallacy

- Levered equity pays $0.5 \times 875 + 0.5 \times 375 = 625$ on average. Discounting at the unlevered 15% gives $625/1.15 = 543$, more than 500. **Wrong**: the discount rate changed.
- Shareholders pay 500 and receive 875 or 375: returns $+75\%$ or -25%, **25%** expected, against $+40\%$ / -10% and 15% without debt. The swing doubled, so did the premium. And $625/1.25 = 500$.

Debt	Equity	r_E	Average cost of capital
0	1000	15%	15%
200	800	17.5%	$0.2 \times 5\% + 0.8 \times 17.5\% = 15\%$
500	500	25%	$0.5 \times 5\% + 0.5 \times 25\% = 15\%$

Debt is cheaper, but equity gets dearer by exactly enough. **Today we put a formula on "exactly enough".**

Recap: MM Proposition I

MM Proposition I. In a perfect capital market, the total value of a firm's securities equals the market value of the cash flows generated by its assets, and is not affected by its choice of capital structure.

- **Why.** Everything the assets produce goes to the security holders, and nothing else does. Same total cash flows, so same total value, whatever the slices.
- Miller's pizza: Yogi Berra asks for eight slices instead of four because he is hungry. Cutting the pizza differently does not make more pizza.
- Galileo without friction: all bodies fall at the same speed. Capital structure matters in practice only through a **friction**: taxes, bankruptcy costs, information.

Recap: homemade leverage

- Investors can build the leverage they want **themselves**, as long as they borrow at the same rate as the firm.
- **More leverage**: borrow 500 and buy the unlevered firm. Same payoffs as levered equity, for 500.

Portfolio	Year 0	Strong	Weak
Buy unlevered equity	-1000	$+1400$	$+900$
Borrow 500	$+500$	-525	-525
Total	-500	$+875$	$+375$

- **Less leverage**: buy the levered firm's debt **and** equity, $500 + 500$. The debt you own cancels the firm's debt: payoffs 1400 or 900, like unlevered equity, for 1000.

Recap: homemade leverage proves MM I

- Firms **U** (no debt, worth V_U) and **L** (debt 500, equity E_L) have the same assets. Suppose their prices differ.
- **U too expensive** (1050 against 1000). Sell U, buy L's debt and shares:

Trade	Year 0	Strong	Weak
Sell U's shares	$+1050$	-1400	-900
Buy L's debt	-500	$+525$	$+525$
Buy L's shares	-500	$+875$	$+375$
Total	$+50$	0	0

- **U too cheap** (990 against $500 + 510$). Sell L's shares, borrow 500, buy U: $+20$ today, 0 next year in both states.

Free money today, nothing changed tomorrow: everyone trades until $V_U = V_L$. The **law of one price** makes MM I hold.

Recap: the market value balance sheet

By MM I, the market value of the firm's assets equals the market value of the securities it has issued:

$$\text{Equity} = \text{Assets} - \text{Debt and other liabilities}$$

- Unlike the accounting balance sheet, **every** asset counts, including intangibles, and all at **market value**.
- Attention: the market value of an asset is often hard, or impossible, to pin down.

Recap: the unlevered cost of capital (L6)

The assets are owned by shareholders and lenders together, so their expected return is the **weighted average** of what each group earns (Chap. 12.5):

$$r_U = \frac{E}{E+D}\, r_E + \frac{D}{E+D}\, r_D \qquad\qquad \beta_U = \frac{E}{E+D}\, \beta_E + \frac{D}{E+D}\, \beta_D$$

- E, D: market values of equity and debt. r_E, r_D: their costs of capital. r_U: the **unlevered** cost of capital, the return of the business itself.
- **Danone**: equity 30 billion at 5%, debt 10 billion at 3.5%, so $r_U = 0.75 \times 5\% + 0.25 \times 3.5\% = 4.6\%$.
- Lecture 6 went **from** r_E **to** r_U, to price a project. Today we go the other way: given the business (r_U) and the leverage, **what must** r_E **be?**

Beginning of lecture 8 content

Lecture 7 showed that leverage does not change what the firm is worth. But debt costs 5% and equity 15%, so how can the average not fall? Because the cost of equity rises exactly enough. Today we see by how much.

Introduction

- **A common argument.** "Debt costs 5%, equity 15%. Replace half of the equity by debt and the average cost of capital drops from 15% to 10%. So why not borrow as much as possible, at least while the debt stays safe?"
- **A banker's version.** In 2010 Josef Ackermann, then CEO of Deutsche Bank, warned that stricter capital requirements (less leverage) would "depress ROE to levels that make investment into the banking sector unattractive".
- Both arguments sound right. Both are wrong for the same reason.

Leverage changes the **risk** of equity, and therefore the return shareholders require. Today we put a number on how much, and show that the saving on cheap debt is given back, cent for cent, through dearer equity.

Outline

A. MM Proposition II: leverage and the cost of equity (Chap. 14.3)

B. The WACC does not move (Chap. 14.3)

C. Levered and unlevered betas (Chap. 14.3)

D. Two fallacies: earnings per share and dilution (Chap. 14.4)

E. Conservation of value (Chap. 14.5)

F. Exercises

A. MM Proposition II: the answer first

The answer first. MM Proposition II. The cost of levered equity is the unlevered cost of capital plus a premium that grows with the debt-to-equity ratio:

$$r_E = r_U + \frac{D}{E}(r_U - r_D)$$

- In words: cost of equity = cost of the business + (debt ÷ equity) × (cost of the business − cost of debt).
- The second term is the **price of leverage**. It is zero without debt and grows with every euro of debt per euro of equity.
- Three steps to get there:
  1. write MM Proposition I as an equation;
  2. compute the return of a portfolio of the firm's debt and equity;
  3. solve for the return of equity.

Step 1: MM I as an equation

- Let E be the market value of the levered equity, D the market value of the debt, U what the equity would be worth if the firm had no debt, and A the market value of the firm's assets.
- MM Proposition I says that the securities are worth the assets, with or without leverage:

$$E + D = U = A$$

- Our entrepreneur: $500 + 500 = 1000$.
- Read the first equality through homemade leverage: holding **both** the debt and the levered equity gives exactly the cash flows of the unlevered equity (Lecture 7, Step 2 of homemade leverage).

Step 2: the return of the portfolio

- The return of a portfolio is the weighted average of the returns of what is in it (Lecture 2). A portfolio holding all the debt and all the equity has weights $E/(E+D)$ and $D/(E+D)$, and earns the unlevered return:

$$\frac{E}{E+D} R_E + \frac{D}{E+D} R_D = R_U$$

- R_E, R_D, R_U are the **realised** returns of levered equity, debt and unlevered equity.
- **Check with the entrepreneur**, weights one half each:
  - strong economy: $0.5 \times 75\% + 0.5 \times 5\% = 40\%$
  - weak economy: $0.5 \times (-25\%) + 0.5 \times 5\% = -10\%$

  Exactly the returns of unlevered equity.

Step 3: solve for the return of equity

- Multiply both sides by $(E + D)/E$ and move the debt term across:

$$R_E = R_U + \frac{D}{E}(R_U - R_D)$$

- The return on levered equity is the return without leverage **plus an extra kick**. When the business does well ($R_U > R_D$) the kick is positive; when it does badly ($R_U < R_D$) the kick is negative. The size of the kick is D/E.
- The equation holds for realised returns, so it holds for **expected** returns too. Replace R by r:

MM Proposition II (Chap. 14.3): $r_E = r_U + \frac{D}{E}(r_U - r_D)$. The cost of levered equity increases with the market-value debt-to-equity ratio.

MM II with the entrepreneur's numbers

- Unlevered cost of capital $r_U = 15\%$, debt cost $r_D = 5\%$.

Debt D	Equity E	D/E	$r_E = r_U + \frac{D}{E}(r_U - r_D)$	Lecture 7 said
500	500	1	$15\% + 1 \times (15\% - 5\%) = 25\%$	25%
200	800	0.25	$15\% + 0.25 \times (15\% - 5\%) = 17.5\%$	17.5%

- Both numbers match what we found last week by looking at the cash flows state by state. Now we have them **without** listing the states.
- The formula also says what happens at **any** level of debt: with 750 of debt ($D/E = 3$), $r_E = 15\% + 3 \times 10\% = 45\%$, the quiz answer of Lecture 7.

B. The WACC does not move: the answer first

The answer first. In a perfect market the weighted average cost of capital is the unlevered cost of capital, whatever the mix (Chap. 14.3):

$$r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D = r_U = r_A$$

- This is Step 2 read with expected returns: the portfolio of all the firm's securities has the risk of the assets, so it earns the return of the assets.
- In Lecture 6 we called this the **pre-tax WACC**, and the WACC had a $(1 - \tau_C)$ on the debt term. Here there are no taxes, so the two coincide.
- Consequence: the enterprise value is the free cash flow discounted at the WACC (Chap. 9). The cash flows do not depend on financing, and neither does the WACC, so **neither does the value of the firm**. That is MM I again.

The WACC at four debt levels

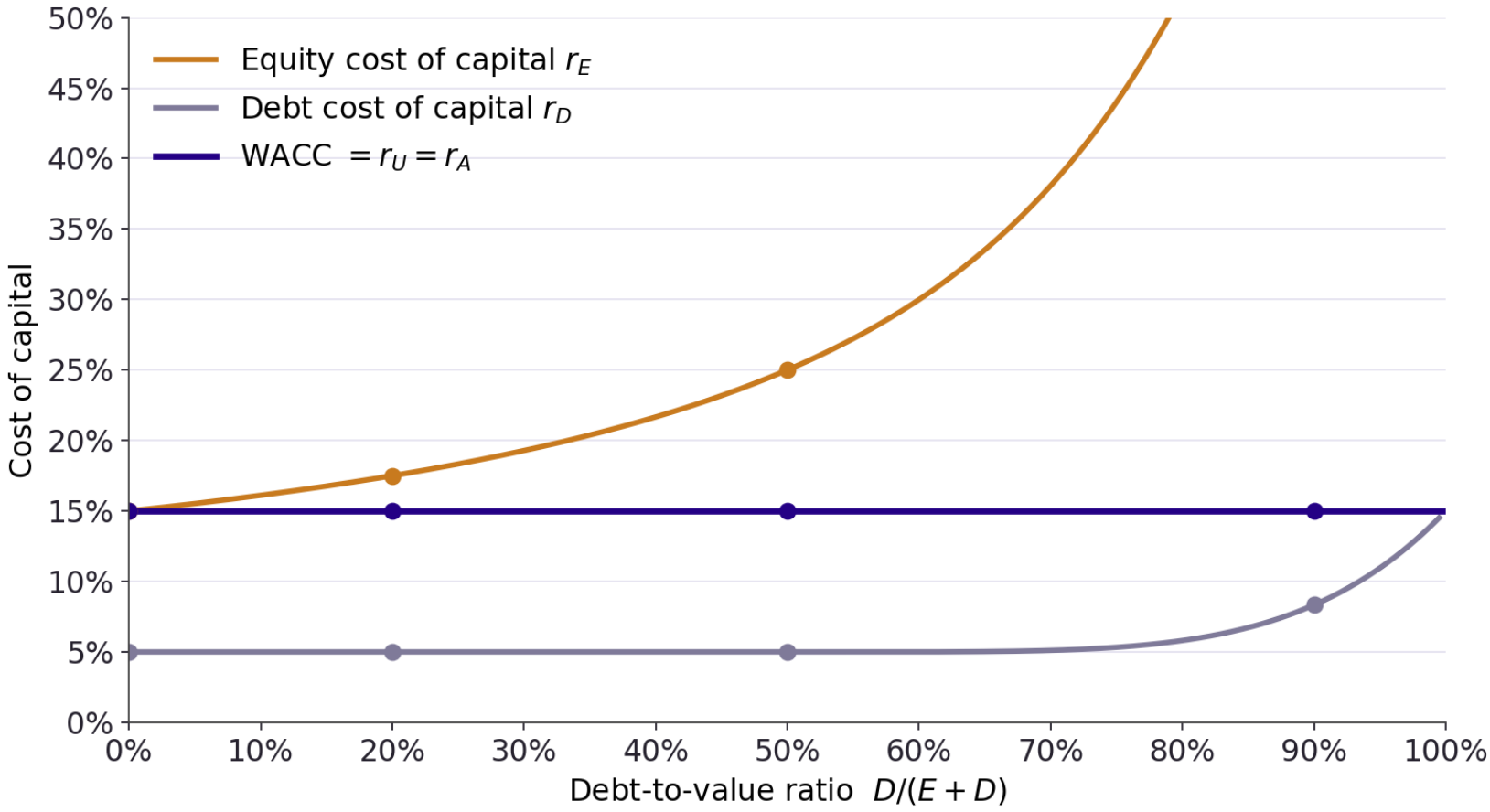
- The entrepreneur's project with more and more debt (Figure 14.1b):

E	D	r_E	r_D	$r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D$
1000	0	15.0%	5.0%	$1.0 \times 15\% + 0 \times 5\% = 15\%$
800	200	17.5%	5.0%	$0.8 \times 17.5\% + 0.2 \times 5\% = 15\%$
500	500	25.0%	5.0%	$0.5 \times 25\% + 0.5 \times 5\% = 15\%$
100	900	75.0%	8.3%*	$0.1 \times 75\% + 0.9 \times 8.3\% = 15\%$

- The cost of equity climbs steeply, but more weight goes on the cheap source: the average stays at **15%**.

* With 900 of debt the firm defaults in the weak state: lenders are promised 16.67% but expect 8.33%.

The WACC in a picture



- As the share of debt rises, **both** the equity and the debt become riskier and dearer. Because more weight goes on the cheaper debt, the average does not move.

Example: NRG reduces its leverage

- **Example 14.5.** NRG Energy has a debt-to-equity ratio of 2, a debt cost of 6% and an equity cost of 12%. With $D/E = 2$, debt is two thirds of the firm and equity one third.
- **Unlevered cost of capital** (and WACC):

$$r_U = \tfrac{1}{3} \times 12\% + \tfrac{2}{3} \times 6\% = 8\%.$$

- NRG issues equity to repay debt until $D/E = 1$. Safer debt now costs 5.5%. **New cost of equity**: $r_E = 8\% + 1 \times (8\% - 5.5\%) = 10.5\%$.
- **New WACC**: $0.5 \times 10.5\% + 0.5 \times 5.5\% = 8\%$. Unchanged.
- If NRG repaid all its debt, its equity would cost $r_U = 8\%$: the WACC again.

Less leverage: cheaper equity, dearer average? No. The WACC, the cash flows and therefore the **enterprise value** are the same in all three cases.

Back to EBS: is debt better than equity?

- Dan Harris can raise the 50 million at 15% (equity) or 6% (debt). With perfect capital markets:
  - borrowing at 6% raises the cost of equity above 15%, by MM II;
  - the **WACC does not change**, so the NPV of the expansion does not change;
  - the value of the firm and its share price do not depend on the choice.
- A tempting objection: "if the debt is risk free, there is no chance of default, so the risk of equity does not change". **Wrong**: the entrepreneur's debt was risk free and the equity still went from 15% to 25%. Leverage concentrates the **business** risk on fewer euros of equity.
- The same holds with any number of securities. **Example 14.6**: 440 of equity at 18.18%, 500 of debt at 5% and 60 of warrants at 75% give $0.44 \times 18.18\% + 0.5 \times 5\% + 0.06 \times 75\% = 15\%$.

C. Levered and unlevered betas

The answer first. Betas obey the same rule as expected returns (Chap. 14.3):

$$\beta_U = \frac{E}{E+D}\,\beta_E + \frac{D}{E+D}\,\beta_D \qquad\qquad \beta_E = \beta_U + \frac{D}{E}\,(\beta_U - \beta_D)$$

- β_U is the **unlevered** (or asset) beta: the market risk of the business, as if it had no debt. β_E is the **levered** (or equity) beta: business risk plus the risk added by leverage. β_D is the debt beta.
- Why the same rule? The beta of a portfolio is the weighted average of the betas in it (Lecture 4), and the portfolio of all the firm's securities has the risk of the assets.
- When a firm changes its capital structure without changing its investments, β_U **stays put and** β_E **moves.**

If the debt is risk free

- With risk-free debt, $\beta_D = 0$ and the second formula simplifies to
$$\beta_E = \beta_U \times \frac{E+D}{E} = \beta_U \left(1 + \frac{D}{E}\right).$$
- Intuition: each euro of equity controls $(E+D)/E$ euros of assets, so it carries that many times the business risk.
- **Example.** A firm with a debt-to-equity ratio of 2 has an equity beta of 1.7 and risk-free debt. Its asset beta is $1.7/(1+2) = 0.57$.
- A firm in the same industry with a debt ratio $D/(D+E) = 0.5$, that is $D/E = 1$, would have an equity beta of $0.57 \times (1+1) = 1.13$.
- The relation between leverage and equity betas is due to Hamada (*Journal of Finance*, 1972).

Example: CVS raises its leverage

- **Example 14.7.** The drug retailer CVS has an equity beta of 0.8 and a debt-to-equity ratio of 0.1. Its debt beta is zero.
- **Unlever**: equity is $1/1.1$ of the firm, so

$$\beta_U = \frac{1}{1.1} \times 0.8 + \frac{0.1}{1.1} \times 0 = 0.73.$$

- CVS borrows until $D/E = 0.5$, with the debt still risk free. **Relever**:

$$\beta_E = 0.73 + 0.5 \times (0.73 - 0) = 1.09.$$

- The equity beta, and with it the cost of equity, rises with leverage. If the debt had taken on some market risk ($\beta_D > 0$), the equity would bear less of it and β_E would rise by less.

Cash as negative debt

- Cash and risk-free securities on the balance sheet are the opposite of leverage: they make the equity **safer** than the business. Measure leverage with **net debt** = debt − cash.
- **Example 14.8.** Cisco, March 2022: market capitalisation 230 billion, debt 12 billion, cash 25 billion, equity beta 0.95, debt beta about zero.
- Net debt $12 - 25 = -13$ billion. Enterprise value $230 - 13 = 217$ billion.
- Unlevered beta, using net debt in place of D:

$$\beta_U = \frac{230}{217} \times 0.95 + \frac{-13}{217} \times 0 = 1.01.$$

- With a risk-free rate of 2.75% and a market risk premium of 5%, the cost of capital of Cisco's business is $2.75\% + 1.01 \times 5\% = 7.8\%$. Because of its cash, Cisco's shares are **less** risky than its business.

The cost of capital of an industry

- Industry average betas are often used, because the firm is not listed or because an average over several firms is a better estimate than one beta.
- **Attention:** firms in the same industry have different leverage. Average the **unlevered** betas, never the equity betas.
- Steps:
  1. Identify a sample of comparable firms.
  2. For each, find the equity beta, the debt-to-equity ratio and, if possible, the debt beta.
  3. Unlever each firm's beta.
  4. Average the unlevered betas.
  5. Relever the average to the target firm's own debt-to-equity ratio.

Beta of an industry: example

- Four listed firms in the same industry, all with risk-free debt:

Firm	Equity beta	D/E	Asset beta $= \beta_E/(1 + D/E)$
A	1.1	23.31%	0.892
B	1.3	44.35%	0.901
C	1.3	2.15%	1.273
D	0.8	3.03%	0.776
Aver-age			**0.960**

- Firm P is not listed, operates in the same industry and has $D/E = 0.2$. What is its equity beta?
- Relever the average: $\beta_E = 0.960 \times (1 + 0.2) = 1.15$.

D. Fallacy 1: leverage raises EPS

The answer first. Leverage raises **expected** earnings per share (EPS), but it also makes EPS riskier. The extra EPS is the pay for the extra risk, so the share price does not move (Chap. 14.4).

- **Levitron Industries (LVI)**: all-equity, expected EBIT of 10 million next year, 10 million shares at 7.50. No taxes, so earnings = EBIT and

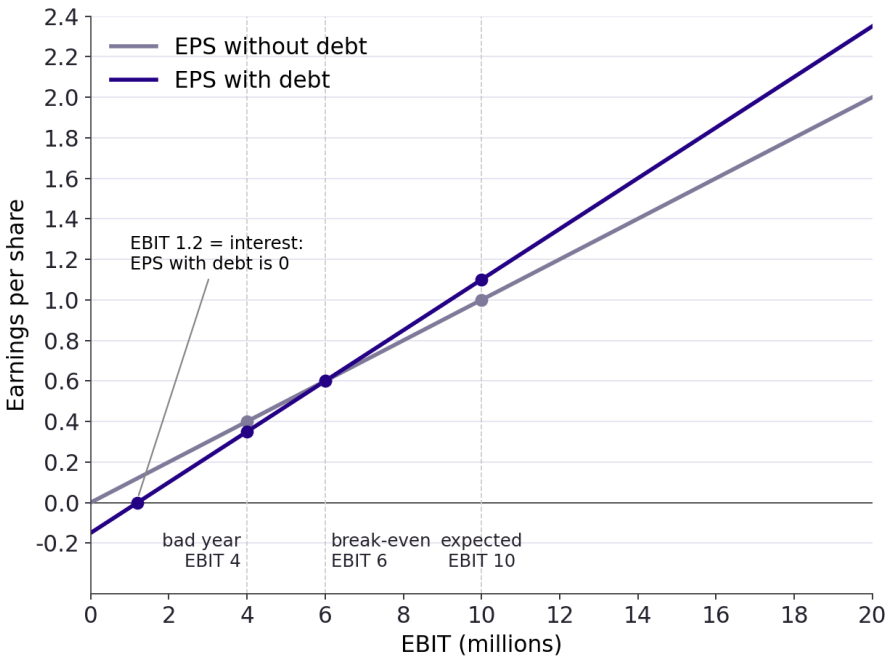
$$\text{EPS} = \frac{10}{10} = 1.00.$$

- LVI borrows 15 million at 8% and buys back 2 million shares at 7.50. Interest: $15 \times 8\% = 1.2$ million. Earnings after interest 8.8 million, shared among 8 million shares:

$$\text{EPS} = \frac{8.8}{8} = 1.10.$$

- EPS up 10%. Shouldn't the share price go up too?

Step 1: EPS in a bad year



- Suppose EBIT is only **4 million**. Without debt EPS is $4/10 = 0.40$. With debt it is $(4 - 1.2)/8 = 0.35$: **lower**.
- The two lines cross at EBIT $= 6$ million. Below it leverage hurts; below 1.2 million (the interest) EPS turns negative.
- The levered line is **steeper**: the same swing in EBIT gives a bigger swing in EPS. Higher average, higher risk.

Step 2: the price does not change

- **Example 14.9.** Suppose LVI's EBIT does not grow and all earnings are paid out. The shares are then a perpetuity, and the price tells us the unlevered cost of capital:

$$7.50 = \frac{1.00}{r_U} \qquad \Rightarrow \qquad r_U = 13.33\%.$$

- After the recap, equity is worth $75 - 15 = 60$ million (MM I), so $D/E = 15/60 = 0.25$. By MM II,

$$r_E = 13.33\% + 0.25 \times (13.33\% - 8\%) = 14.66\%.$$

- Price per share with the new EPS and the new required return:

$$P = \frac{1.10}{14.66\%} = 7.50.$$

Higher EPS, higher required return: the two cancel. EPS, the P/E ratio and the return on equity (ROE) all depend on leverage, so they **cannot be compared** across firms with different capital structures. Analysts prefer EV/EBIT.

Aside: banks and the ROE fallacy

- Banks are extremely levered: international rules allow common equity to be as little as **3%** of total funding (Basel III), against more than 50% for a typical non-financial firm. With such a thin cushion, a small drop in asset values means insolvency: a key ingredient of the 2008 crisis.
- When regulators asked for more equity, bankers (Ackermann, slide 13) replied that it would lower their **return on equity** and make bank shares unattractive.
- MM: lower leverage lowers the **average** ROE and lowers its **risk** by the same token. In a perfect market, shareholders are no worse off.
- Bank leverage is attractive only because of imperfections, mostly government guarantees. The gains come largely at the taxpayer's expense.

Fallacy 2: issuing equity dilutes

The answer first. Selling new shares at a fair price does not dilute value: the cash raised adds to the assets exactly as much as the new shares add to the share count.

- **Jet Sky Airlines**: no debt, 500 million shares at 16: worth 8 billion, expansion included. It needs 1 billion of planes and sells **62.5 million new shares at 16**.

In millions	Before the issue	After the issue
Cash	0	1000
Existing assets	8000	8000
Total value	**8000**	**9000**
Shares outstanding	500	562.5
Value per share	16.00	16.00

- More shares, but also more assets: $9000/562.5 = 16$. Gains or losses come only from the **NPV of what the money buys**. Announcement price drops are an information effect, a friction.

E. Conservation of value

Conservation of value. With perfect capital markets, financial transactions neither add nor destroy value. They **repackage** risk, and therefore return (Chap. 14.5).

- John Burr Williams wrote it down in 1938, twenty years before MM: the value of a firm is the present value of everything it will ever pay out, and that does not depend on whether the payments are called interest or dividends. He compared it to the conservation of matter or energy in physics.
- The principle goes far beyond debt versus equity. **Any** financial deal that seems to create value is either too good to be true or exploits a market imperfection. Before believing in the value, **name the imperfection**.
- Modigliani and Miller's real legacy: the role of financial policy is to deal with, and sometimes exploit, the frictions of real markets.

What this means for a CFO

- **Value comes from investments, not from financing.** Positive-NPV projects make the pizza bigger; the capital structure only cuts it into slices.
- It is far easier to create or destroy value with investment decisions than with financing decisions.
- So why do CFOs rank capital structure first (Lecture 7)? Because real markets have frictions, and each friction gives financing a role:
  - **taxes**: interest is deductible, dividends are not (Lecture 9);
  - **financial distress** and the costs of going bankrupt;
  - **managers' incentives** and what financing choices **signal** to investors (Lectures 10 and 11).

MM in a perfect market is the benchmark. Every reason for leverage to matter is a departure from one of its five conditions.

F. Exercise 1: cost of capital and leverage

- Amalgamated Products (Am.Pr.) owns three activities: Food (40% of the value of the firm), Electronics (30%) and Chemicals (30%).
- Amalgamated Products doesn't pay tax. To estimate its cost of capital Amalgamated has identified three competitors:

	Beta of the stock	Debt/(Debt+Equity)
United Foods	0.8	0.4
General Electronics	1.6	0.2
Associated Chemicals	1.2	0.3

- We assume that the debt of the firm is risk-free.
  - (a) What are the asset betas of each division of Amalgamated?
  - (b) The debt ratio of Amalgamated is D/(D+E)=0.4. What is the beta of the stock of Amalgamated?
  - (c) With $r_f = 7\%$ and $r_M = 15\%$, what is the cost of capital of each division?
  - (d) What is the WACC of Amalgamated?

Exercise 2: reducing leverage

- Infonian AG is a very leveraged firm with a debt to equity ratio of 150%. The bank does not consider the debt as risky and asks for an interest rate of 5% ($\approx r_f$). The equity holders require a return of 20%. The company does not pay tax.
  - What is the WACC of the firm?
  - What is the cost of equity if Infonian issues equity and reduces its debt to achieve a debt to equity ratio of 50%?

Exercise 3: increasing leverage

- Mr. Goodman the CFO of Perkscom comes back thoughtful from the shareholder meeting of his firm. The investment fund who owns 10% of the capital of Perkscom has announced to the CFO Mr. Goodman at the annual shareholder meeting that it expects an earnings yield of at least 20% for the current year, otherwise it would take away the stock options for the managers of the firm. Because Mr. Goodman counts on his stock options to repay the mortgage for his villa, he is a bit worried and thinks about what he can do.
- The cash flow of Perkscom is equal to the EBIT of 10 million for this year with annual growth rate of 10% per year. The firm has no leverage. The accounting value of its equity is 80 million and its stock market capitalisation is 100 million.
- The firm could easily raise debt at an interest rate of 5%. Perkscom does not pay tax (tax rate $= 0$) and distributes all its cash flows as dividends to shareholders.

Exercise 3: increasing leverage (con't)

1. If the market anticipates the perpetual growth rate of the EBIT of 10%, what should be the cost of capital of Perkscom?
2. Suppose that with a more efficient reorganisation of its production, Perkscom can increase its cash flows by 50% from 10 million to 15 million at the end of the year with the same perpetual growth rate as previously. What is now the value of the firm and the return on equity and earnings yield?
3. Mr. Goodman understands that with operational measures, he cannot achieve the earnings yield required by the investment fund. He decides to increase the leverage to the debt-to-equity ratio of 3:1 without changing the assets of the company (e.g., we are in the initial setting and not the one of 2.) and without increasing the cashflows of the company. Compute the amount of debt to be raised and describe briefly the necessary transactions.

Exercise 3: increasing leverage (con't 2)

4. What will be the WACC after the transaction and what will be the cost of equity?
5. What will the earnings yield and return on equity be?
6. What is the lowest level of leverage that can achieve the earnings yield required by the investment fund?
7. Has the investment fund chosen a good performance measure? Should it be satisfied by the measures taken by Mr. Goodman?

Next time: taxes break the symmetry

- Everything today rested on condition 2 of a perfect market: **no taxes**. In the real world, firms pay tax on their profits **after** interest, and dividends come out of **after-tax** profit.
- **Macy's, 2014**: EBIT 2.8 billion, interest 400 million, tax rate 35%. The interest reduced taxable profit by 400 million, so Macy's paid $35\% \times 400 = 140$ million **less tax** than an unlevered Macy's.
- That 140 million a year is the **interest tax shield**: more pizza for investors, because less goes to the tax authority.
- Lecture 9 (Chap. 15): a levered firm is worth an unlevered firm **plus** the present value of its tax shields, and the WACC now **falls** with leverage.

How it all connects

1. **MM I as an equation**: $E + D = U = A$. Holding all the debt and all the equity is holding the business.
2. **The return of that portfolio** is the return of the business, so solving for the equity return gives **MM II**: r_E rises with D/E.
3. **The WACC is flat** at the unlevered cost of capital: the saving on cheap debt is returned through dearer equity. Enterprise value does not depend on financing.
4. **Betas follow the same rule.** Unlever comparables before averaging, relever to the target; treat cash as negative debt.
5. **EPS, ROE and P/E rise or fall with leverage** only as compensation for risk; selling shares at a fair price dilutes nothing.
6. **Conservation of value**: when a financial deal seems to create value, look for the friction.

What to remember

- **MM II**: $r_E = r_U + \frac{D}{E}(r_U - r_D)$. Debt is cheaper than equity, but leverage raises the cost of equity, even with risk-free debt.
- **Without taxes, the WACC equals the unlevered cost of capital**, whatever the capital structure. Leverage does not lower the discount rate for the firm's projects.
- **Same formulas with betas**: $\beta_E = \beta_U + \frac{D}{E}(\beta_U - \beta_D)$; with risk-free debt, $\beta_E = \beta_U(1 + D/E)$.
- **Unlever before you average** comparables, and use net debt.
- **EPS, P/E and ROE are not comparable** across capital structures. Fair-price equity issues do not dilute value.
- **Value comes from investments.** Financing matters only through frictions.

Quiz

- Why is debt cheaper than equity?
- Why does the WACC not fall with leverage when there are no taxes?
- A firm without debt has a beta of 1. What is its equity beta if it takes on debt equal to 50% of the value of its assets? (The debt beta is zero.)
- LVI's expected EPS rose from 1.00 to 1.10 after its recap. Why did the share price stay at 7.50?

Quiz: answers

- **Debt is cheaper** because lenders are paid first and carry little of the business risk. The equity carries the rest, so it must be dearer.
- **The WACC does not fall** because every euro of cheap debt makes the remaining equity riskier. By MM II the cost of equity rises by exactly enough to keep the weighted average at the unlevered cost of capital.
- **Equity beta.** Debt is 50% of the assets, so $D/E = 1$ and $\beta_E = \beta_U(1 + D/E) = 1 \times (1 + 1) = 2$.
- **LVI.** The required return on equity rose from 13.33% to 14.66% along with the EPS, because the EPS became riskier. $1.10/14.66\% = 7.50$: the two effects cancel.